PART 6 | STATISTICAL ANALYSIS

Statistical Analysis Report

Descriptive statistics, association analysis, and five prespecified hypothesis tests

Methods and interpretation

The analysis uses observation units suited to each question: one row per patient for age, first encounter per patient for age relationships, admitted-patient records for severity versus stay, department summaries for workload and satisfaction, and 12 monthly observations for receipts versus patient volume. Statistical significance is evaluated at alpha = 0.05. Holm adjustment is applied across the five tests to control family-wise error.

Measure	N / grain	Key results
Age	4,443 patients	Mean 45.52; median 46; Q1 23; Q3 68; P10 9; P90 81.
Length of stay	5,000 appointments	Median 0 days; P90 2 days. Zero values include non-admissions.
Billed amount	5,000 appointments	Mean 154,226.69; median 153,724; SD 83,630.75.
Payment amount	5,000 appointments	Mean 138,910.26; median 128,223.50.
Patient satisfaction	1,756 rated records	Mean 2.925; median/mode 3; SD 1.386.
Department workload index	200 departments	Mean/median 50.25; roster-based proxy.

Association results

Relationship	Coefficient	p-value	Interpretation
Age vs first-visit admission	r = 0.0255	0.0893	Very weak; not significant at 0.05.
Age vs first-visit billing	rho = -0.0067	0.6568	No meaningful monotonic relationship.
Severity vs stay among admitted	rho = -0.0170	0.6843	No observed monotonic association.
Department workload vs satisfaction	rho = -0.0245	0.7310	No observed association; workload is a proxy.
Monthly receipts vs patient volume	rho = 0.9161	0.00003	Strong positive association across 12 months.

Hypothesis tests

Test	Method	Statistic / p	Holm p	Decision
H1 Satisfaction differs across facilities	Kruskal-Wallis	H=14.931; 0.72695	1.00000	Fail to reject H0
H2 Admission rate differs by age group	Chi-square, df=4	chi2=9.752; 0.04482	0.17927	Fail to reject after adjustment
H3 Severity associated with stay	Spearman	rho=-0.01695; 0.6843	1.00000	Fail to reject H0
H4 Workload associated with satisfaction	Spearman	rho=-0.02446; 0.7310	1.00000	Fail to reject H0
H5 Receipts associated with patient volume	Spearman	rho=0.91608; 0.00003	0.00014	Reject H0

The unadjusted H2 p-value is below 0.05, but it does not remain significant after Holm correction across five related tests. The effect is small (Cramer's V = 0.0469), so age-group differences are weak in practical terms. H5 remains significant after adjustment; its strong association is descriptive and based on only 12 months, so it does not establish causality or forecast performance.

Conclusions

The evidence does not support facility differences in satisfaction, an association between severity and stay, or a relationship between the department workload proxy and satisfaction. Monthly receipts and patient volume move together strongly. Interpret all null results in context of proxy quality, incomplete records, and test power.

Statistical caveats

• P-values quantify compatibility with the null under the test assumptions; they do not measure the probability that H0 is true.

• Holm-adjusted p-values are the appropriate decision values when treating these five tests as one inferential family.

• Repeated visits can inflate apparent sample size; the analysis uses deduplicated or aggregated observations where specified.

• Source monetary units are not labeled with a currency, and insurance identifiers do not verify active coverage.

• The receipt-volume test has twelve monthly observations and should not be generalized beyond this cohort.

Reporting standard

The notebook records each test's hypotheses, method, statistic, sample size, effect size where applicable, raw p-value, Holm-adjusted p-value, and decision. The detailed tables should be consulted alongside this summary before any operational policy change.